

H1 Biology Preliminary Examination

Answer Key

Paper 1

Question	Answer	Question	Answer
1	A	16	B
2	B	17	D
3	D	18	B
4	D	19	B
5	A	20	A
6	D	21	D
7	A	22	D
8	C	23	B
9	C	24	B
10	D	25	D
11	A	26	C
12	D	27	C
13	D	28	D
14	B	29	B
15	D	30	C

Paper 2 (Section A)

1. **Fig. 1.1 shows a ribbon diagram of a protein at the tertiary level of protein structure.**
 - (a) **Identify the type of secondary structure labelled X and Y.**
X: alpha helix
Y: beta sheets [1]
 - (b) **Identify a type of protein that would most likely have a 3-D configuration shown above.**
Enzymes / globular proteins [1]
 - (c) **A group of tertiary proteins are twisted and folded such that there is a special functional site (labelled Z in Fig 1.1) within the structure. Name this site.**
Site Z: Active site [1]
 - (d) (i) **Explain the results of the experiment as shown in Fig. 1.2.**
 - Iodine reacts with starch to give a blue black coloration
 - Amylase hydrolyses starch maltose, which no longer reacts with iodine, thus giving a brown coloration
 - The activity of amylase increases with increasing temperature.
 - The higher the temperature, the faster the hydrolysis and earlier is the blue black colouration disappears
 - Thus at 30°C, the blue black coloration disappears after 45s while at 10°C, the blue black colour takes 195s to disappear.

Max [4]

(ii) **What role does Site Z have in the above reaction and how does changes in temperature affect site Z?**

- Site Z is the active site which binds the substrate (starch) and hydrolyses it into maltose
- Increasing temperature increases the kinetic energy of the enzyme and substrate molecules thus resulting in successful collision between the molecules
- Resulting in enzyme-substrate complex formation and thus hydrolysis
- At high temperatures, beyond the optimum temperature, shape of the active site will be lost
- Thus the hydrolysis of starch cannot take place

max [3]

[Total:10]

Question 2

(a) **Write down the genotype and phenotype of the F1 offspring from the cross stated.**

BbNn; phenotype: black and wakeful(non-narcoleptic). [2]

(b) **Two individuals of the F1 were subsequently mated to generate several litters of puppies. Assuming that the two genes are unlinked, perform a genetic cross in the space below to obtain the expected phenotypic ratio in the F2 generation. Show your workings clearly.**

- Parental gametes;
- Punnett square(2m);
- Phenotypic ratio; [4]

(e) **Given that the formula for chi-square below along with a section of the chi-square table, perform a chi-square test to prove that the two genes are unlinked (Show your workings clearly and in a detailed manner).**

- Setting of null hypothesis;
- Summation working;
- Calculated chi-square value of 0.94;
- Locating critical chi-square value of 7.82;
- Conclusion of accepting null hypothesis; [4]

Question 3

(a) **Explain how mouthbrooding behaviour in certain cichlid species might have evolved.**

- Variation in breeding habits
- Unusual genetic-linked behaviour might have arisen out of chance / mutation
- Behaviour allows bearer to survive better and/or reproduce more than other cichlids
- Mouthbrooders ensure a large percentage of their offspring survive till relatively matured.
- Inheritable trait passed down to offspring
- Mechanism of natural selection leading to evolution [4]

	DNA sequence
Species A	A-T-C-T-G-G-A-T-C-A-G-G-G-C-C-C-G-G-A-T-T
Species B	A-T-G-A-G-G-T-T-C-A-G-G-A-C-C-G-G-A-T-T
Species C	A-T-C-A-G-G-T-T-C-A-C-G-A-C-C-G-G-A-T-T
Species D	A-T-C-T-G-G-A-T-C-A-G-G-G-C-C-C-G-G-A-T-A

(b) **In comparison to the DNA sequence in species A, note down the number of nucleotide differences that species B, C, and D have.**

A-B: 5, A-C:5, A-D:1 [1]

- (c) **Based on the DNA sequence presented, construct a cladogram showing the evolutionary relationship between the four species.**

Answer should show A on the same branch as D;
and B on the same branch as C;

[2]

- (d) **Explain three advantages in using molecular homology (especially DNA homology) in this scenario to clarify evolutionary relationships, in comparison to using anatomical or embryological homology.**

- Less subjective as it is quantitative / precise differences;
- Can be used to tell evolutionary time (molecular clock) due to accumulated mutations being regular over long periods of time;
- Organisms might have looked very different and yet diverged recently due to environmental pressures;
- Or be very similar morphologically but have undergone long periods of separation;
- DNA mutation is relatively constant / regular over very long periods of time;
- Number of nucleotide differences could be used as a reliable reference for relation;[3]

4. Restriction enzymes (also known as restriction endonucleases) are used to cut DNA molecules.

- (a) **State the source and natural use of restriction enzymes.**

Source: bacteria (1) natural use: defence against viral infection (1) [2]

- (b) **Explain whether it would be possible to cut this sequence using restriction enzymes B or C.**

- (i) **enzyme B**

Yes because the target site in the plasmid is the same as the restriction sequence for enzyme B [1]

- (ii) **enzyme C**

No, The target site in the plasmid is not the same as the restriction sequence for enzyme C [1]

- (c) **Suggest how bacterial B is possible.**

- plasmid re-annealed without taking up a foreign DNA fragment (1)
- and plasmid is taken up by bacterial cell.(1) [2]

- (d) **Using the plasmid illustrated in Fig. 3.2, explain how you would be able to select for bacterial C, assuming all three types of bacterial cells were present in the same culture medium.**

- culture bacteria in agar medium with ampicillin.
- Bacterial cells A will not survive in the medium as it has no resistance to ampicillin.
- Culture the surviving bacterial cells in medium containing X-gal.
- Bacterial cell B will grow on the agar medium as blue colonies as the β -gal gene is intact.
- Bacterial cell C will grow as white colonies as it is unable to breakdown X-gal, as the β -gal gene is disrupted by gene of interest, hence cannot express the enzyme.

1m per point; max [4]

[Total:10]

Section B

- 5 (a) **Describe phosphorylation, giving examples wherever possible, indicate if this process occurs in photosynthesis, respiration or both.** [4]

- The addition of phosphate group to ADP/ synthesis of ATP from ADP and P_i / addition of phosphate group to organic molecules eg sugars/ protein
- Occurs in both photosynthesis and respiration

Examples

In photosynthesis

- Photophosphorylation occurs on the thylakoid membrane of chloroplast
- Light reactions consist of non cyclic and cyclic phosphorylation

In respiration

- Oxidative phosphorylation occurs on the inner mitochondrial membrane
- Substrate level phosphorylation occurs in the cytosol and matrix of mitochondrion
- In glycolysis: from 1,3 diphosphoglycerate to 3 phosphoglycerate
from 3 phosphoglycerate to pyruvate
- Krebs cycle; from succinyl CoA to succinate

1 mark per point; max 4

- (b) **Explain how the structure of mitochondria allows respiration to proceed.**

- Outer mitochondrial membrane compartmentalizes / separate contents of organelle from cytoplasm for aerobic respiration to occur
- Allows reaction to occur independently from each other
- Inner mitochondrial membrane contains ETC and ATP synthetase to carry out oxidative phosphorylation
- Cristae/ folding to increase surface area
- Enzymes for Krebs cycle and link reaction found in matrix
- Intermembrane space concentrates protons to create a gradient for chemiosmosis to occur
- Membranes not permeable to ions

1 mark per point, max [6]

- (c) **Compare and contrast Krebs and Calvin cycles.**

Similarities

- Involve regeneration of starting compound/ cyclic processes
- Respiration: oxaloacetate regenerate while in PS, it's RuBP
- both do not require light
- both utilize electron/ hydrogen carriers
- Krebs cycle: NAD and FAD; Calvin cycle: reduced NADP/ NADPH

1 mark per point; max 3

Differences:

Krebs	Calvin
Matrix of mitochondria	Stroma of chloroplast
In aerobically respiring cells	In plant cells/ algae/ blue green bacteria
Catabolic reaction	Anabolic reaction
Production of ATP via substrate level phosphorylation	Expenditure of ATP (reduction of GP to TP and regeneration of RuBP)
NAD and FAD as electron acceptors	Reduced NADP/ NADPH –electron donor
Oxidation of substrate (acetyl CoA)	Reduction of substrate (CO_2)
Reactions mainly oxidative/ many NAD formed	Reactions mainly reductive where NADPH is consumed
CO_2 released	CO_2 consumed/ fixed by RuBP
1 cycle – net loss of 2C	1 cycle net gain of 1C
Starting material –oxaloacetate	Starting material - RuBP
Not light dependent	Depends on light indirectly.

1 mark per row of comparative points; max 7 [Total: 20]

(a) Outline the differences between the prokaryotic control of gene expression with the eukaryotic model. [8]

(point-by-point comparison preferred)

Transcription in eukaryotes occurs in nucleus, and translation occurs in the RER / cytosol, whereas transcription and translation is continuous in bacteria;

Prokaryote genes organised in operons where structural genes could be grouped together; Whereas an eukaryotic gene is usually individually controlled by a single promoter;

lac operon is controlled by regulatory elements such as *lacI* coding for the repressor which binds to the operator; whereas eukaryotic transcription is dependent on regulatory sequences termed enhancer and silencer regions proximal or distal to the gene;

Transcription of structural genes is controlled by the availability of glucose in prokaryotes; whereas activators and repressors are molecules in eukaryotes that regulate the transcription process;

Description of activators and repressors' role in transcription;

Post-translational modifications and control in eukaryotes: capping and tailing;

Introns and alternate splicing (2m);

(b) Outline the ways in which mutations could occur. [6]

- Definition of mutation as a change in nucleotide sequence for a gene (or change in chromosome structure / number);
- Conditions that promote rate of mutation - exposure to carcinogen, ultraviolet radiation, etc;
- Error occurring during replication;

Single nucleotide mutation (max 2m)

- Substitution - replacement of a nucleotide / nucleotides with different nucleotide / nucleotides
- Deletion - removal of nucleotide / nucleotides
- Insertion - addition of nucleotide / nucleotides

Multi-nucleotide mutation (1m)

- Inversion - reversing the sequence for a particular stretch of nucleotides
- Translocation - removal of stretch of nucleotides and inserting elsewhere in the genome;

Frameshift mutation - change in nucleotide sequence (for example, addition / deletion of one or two nucleotides), causing subsequent codons to be read differently, due to the triplet coding;

Chromosomal mutation (2m)

- Translocation - where part of a chromosome is broken off and fused with another chromosome;
- Aneuploidy - change in chromosomal number due to non-disjunction;

DNA repair - most DNA mutations are repaired, but if not repaired before division into two daughter cells, mutations might become permanent (for the two cells);

(c) Describe how mutations might lead to cancer. [6]

Growth of cancer (multi-step)

- Normal cells communicate with each other and regulate each other's proliferation, dividing only when required;
- Communication depending on protein signals encoded by genes that regulate cell growth and division;
- Small number of mutations initially, affecting genes that regulate cell division;
- proto-oncogenes, which might be converted into oncogenes when mutation occurs;
- oncogenes encode for products that encourage the cell to grow and divide;

- or tumour suppressor genes, such as p53;
- p53 encodes for genes that prevent the mutated cells from entering the cell cycle, or to promote apoptosis in such cells;
- Mutations might causing the p53 they encode to become dysfunctional;
- These harmful mutations might accumulate over several years;
- Leading to uncontrolled cell proliferation;
- A small group of proliferating cancer cells might influence surrounding cells to divide as well;
- If genes coding for adhesion proteins are mutated, cancer cells might move to another region of the body and grow there;